

TABLE II
De-soldering Stations for Through-hole (Vacuum Pump Type)

Specifications	Manufacturer					
	OK International	GOOT, Japan make	Weller	HAKKO	Srinivasa Electronics	Metcal
Model	MFR-1350	TP-200	WR 2002	Model-474	PLUTO 777	MFR-1351
Temperature controller (Yes/No)	Not required. Works on better Smartheat technology	Yes	Yes	Yes	Yes	Yes
Power consumption (W)	110	170-200	300 (desoldering iron 80)	100	200	110
Temperature range (°C)	Not applicable. Works on Smartheat technology	250-450	50-450	380-450	180-480	—
Temperature stability (±°C)	±1.1 (Ideal tip)		±2		1%	±1.1 in still air
Nozzle to ground resistance (Ω)	<2	<2		<2	< 2	<2
Nozzle to ground potential (mV)	<2	<2		<2	0.2	<2
Vacuum generator type	Inbuilt pump		Inbuilt pump with periodic duty 30/30s	Double cylinder type	Diaphragm	Internally
Vacuum pressure (N/m ²)	70,000	79,993.2	70,000	80,000	79,993.2	70,000
Heater type and integration (Integrated in the handpiece or separate)	Smartheat-based heater. Never requires tip temperature verification or calibration of the system	Ceramic heater		Ceramic heater -integrated in the handpiece	Separate	Integrated in the handpiece
ESD-safe (Yes/No)	Yes	Yes	Yes	Yes	Optional	Yes
Dimensions (mm)	170×200×152.5	95×256×205	273×235×102	165×135×260	—	170×200×152.5
Weight (kg)	—	Total weight of unit: 2.9, De-soldering iron: 0.3	6.7	3.8	4	
Cord length (m)	1.52	1.5	1 (approx.)	1.2	1	1.83

TABLE III
Hot-Air Rework Systems

Specifications	Manufacturer					
	OK International	GOOT, Japan	Weller	HAKKO	Srinivasa Electronics	METCAL
Model	HCT-900	XFC-100		FR-803B	PLUTO 888	HCT-900-21
Temperature controller (Yes/No)	Yes	Yes	Yes	Yes	Yes	Yes
Power consumption (W)	320	380-420	300, Hot air iron-100	30	200	320
Temperature range (°C)	100-500	150-450	50-550	100-500	180-500	100-500
Airflow (L/min)	5-20	5-20	10	5-20	24	6-25
Dimensions (mm)	210×170×140	227×107×177	273×235×102	160×145×230		210×170×140
Weight (gm)	115	4600	6700 (approx.)	5000	4000	4700
Cord length (m)	1.22	1.1	1 (approx.)	1.2	1	1.83
ESD-safe (Yes/No)	Yes	Yes	Yes	Yes	Optional	Yes

types of joints. For large components and joints you would need more power, and you would not like to buy another station for that kind of work.

Temperature range. Different solder joints have different nozzle temperature requirements for de-soldering. Also, different solders have different melting temperatures

(see Table I). So it makes sense for the de-soldering station to be temperature-controlled type. Check out the range of joints you would be de-soldering and select a station with